Dental Treatment of Patients with Acute & Chronic Pulmonary Diseases

OMFS 512: Integrated Biomedical and Clinical Sciences

Rashidah Wiley, DDS, FAAOMP, D-ABOMP

Learning Outcomes & Competency Statements

Learning Outcomes

1. Discuss the clinical features associated with upper respiratory tract infections.
2. Compare and contrast sinusitis of the nasal cavity and odontogenic sinusitis.
3. Discuss dental treatment consideration and modifications associated with pulmonary disorders.
4. Review the difference between obstructive sleep apnea and central sleep apnea.

Competency statements met

#1.2,1.3, 5.1

Topics

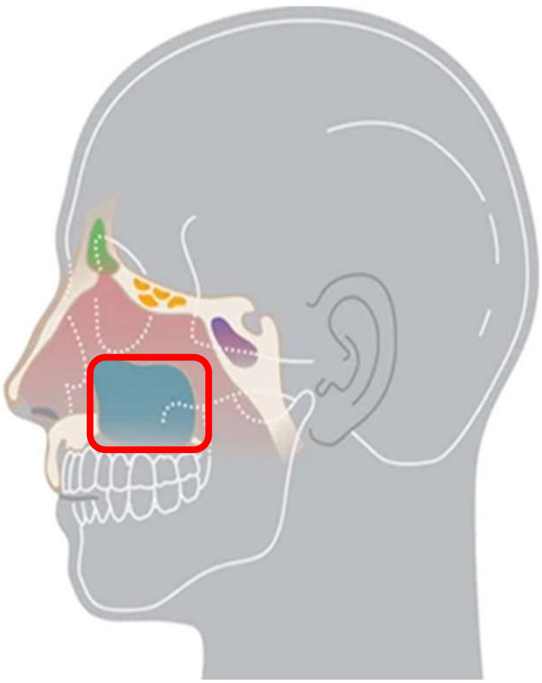
- Sinus Inflammation
  - Allergic rhinitis
  - Sinusitis
- Lung Infections
  - Pneumonia
  - Tuberculosis
- Obstructive Lung Disorders
  - Asthma
  - Bronchiectasis
  - Acute bronchitis
  - Chronic obstructive pulmonary disease (COPD)
  - Emphysema
- Lung Cancer
- Smoking and Tobacco Cessation
- Sleep Apnea

Allergic Rhinitis

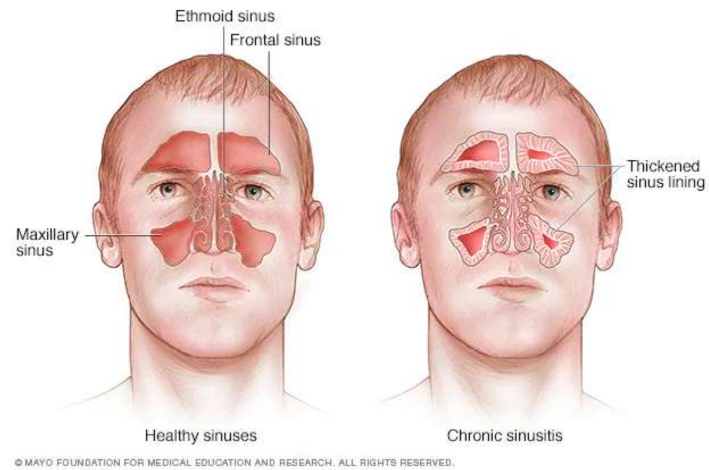
- Seasonal allergies "hay fever“
  - IgE-related type I hypersensitivity reaction
- Allergic reaction to household and environmental allergens
- Inflammation and swelling of mucosal tissues
  - Eyes, nose, and paranasal sinuses
  - "Watery eyes"
  - Clear discharge form the nose
  - Nasal polyps

Allergic Rhinitis

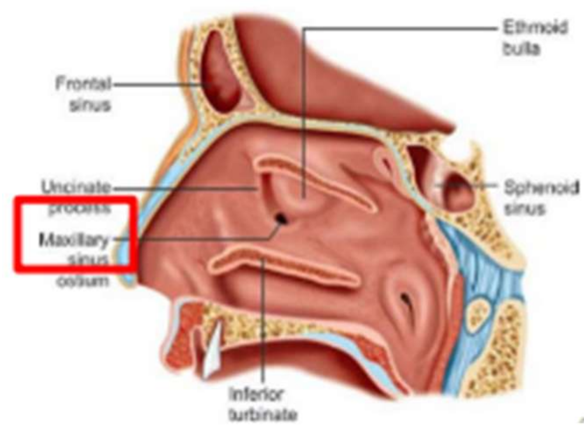
- Sinus pressure from the inflammation can mimic dental pain
  - Pain occurs in the posterior maxillary teeth
- Xerostomia
  - Mouth breathing
    - Gingival hyperplasia, narrowing of the facial profile
    - Class II occlusion/overbite
  - Antihistamine and corticosteroid medications
    - Decreased salivary flow rate
    - Alteration to the oral flora
- Complications with sleep apnea



Sinusitis



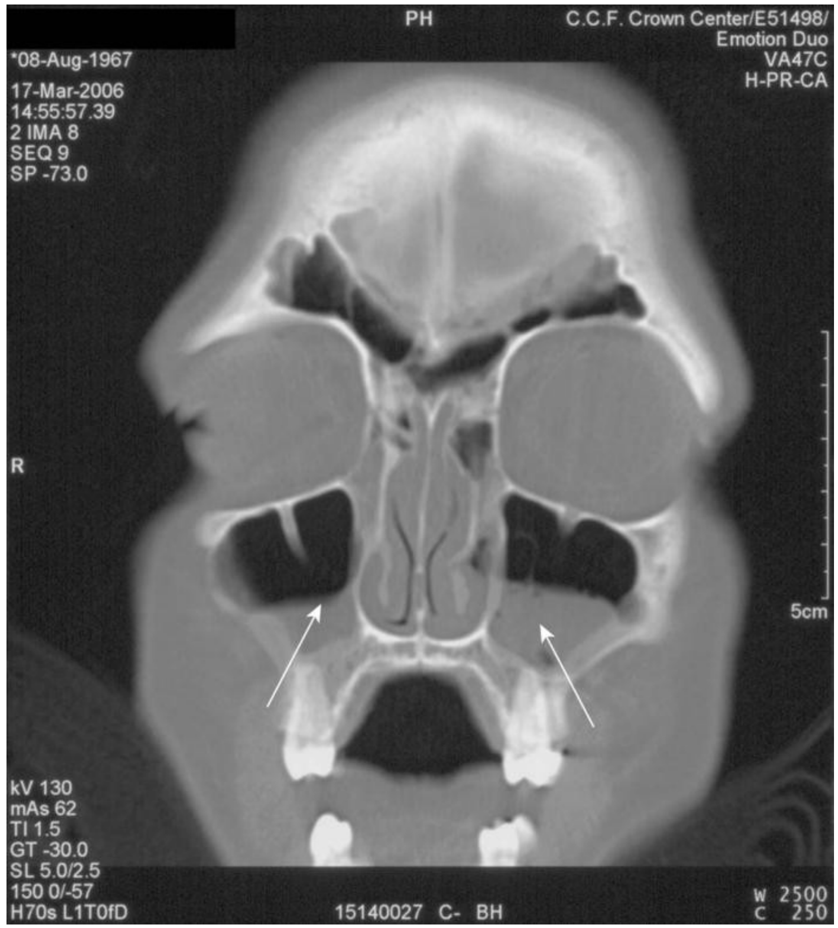
- Rhinosinusitis
- Inflammation of the mucosal lining one or multiple of the paranasal sinuses
- Types of Sinusitis
  - Acute
  - Subacute
  - Chronic
  - Recurrent acute
  - Odontogenic rhinosinusitis



Sinusitis of Non-Odontogenic Origin

Acute Sinusitis	Subacute Sinusitis	Chronic Sinusitis	Recurrent Acute Sinusitis
Symptoms last > 4 weeks	Symptoms last 4-12 weeks	Symptoms last for 12+ weeks	Symptoms last for 12+ weeks, lacks symptoms of sinusitis in between episodes
Viral upper respiratory tract infection Secondary bacterial infection		Multifactorial: allergy, bacterial, and fungal infections	
Purulent nasal discharge Nasal obstruction/congestion Facial pain and pressure **Dental pain**		Purulent nasal discharge Nasal obstruction/congestion Facial pain and pressure Olfactory dysfunction Headache Fatigue **Dental pain** **Xerostomia** **Gingivitis** **Halitosis**	
Self-limiting Medications: Pain medication, antipyretics, and decongestants		Nasal irrigation, corticosteroids, decongestants, antihistamines, antibiotics, and antifungals Functional endoscopic sinus surgery	

3-D Image: Thickening of the Sinus Mucosa



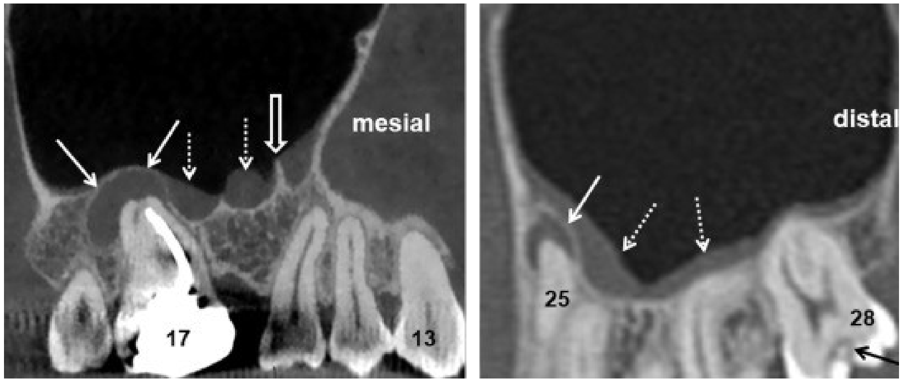
https://www.clevelandclinicmeded.com/medicalpubs/diseasemanagement/allergy/rhino-sinusitis/

Odontogenic Sinusitis

- Sinusitis induced by an odontogenic source
    1. Odontogenic infection of the posterior maxillary teeth
    2. Trauma or surgery involving the posterior maxillary teeth
    3. Tooth/foreign body displacement into the maxillary sinus
- Polymicrobial infection in of the sinus
    - Perforation of the maxillary sinus allows bacteria from the oral cavity to enter the sinus

Odontogenic Sinusitis

- **Symptoms are unilateral**
  - Sinus pain
  - Tooth pain
  - Pain over the cheek bone
  - Postnasal drip
  - Congestion
  - Nasal discharge



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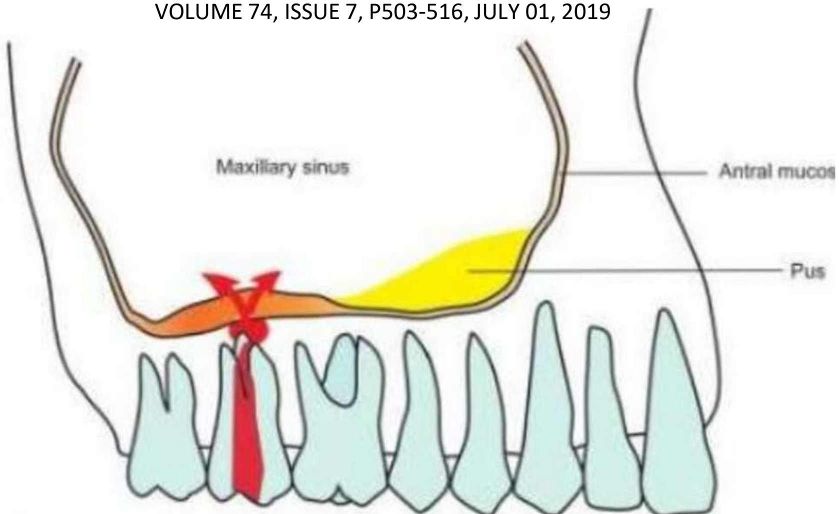

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Management of Sinusitis

- Antibiotics + topical/oral decongestants
- Nasal rinses
- Corticosteroids
- Surgery
  - Deviated septum
  - Ostiomeatal disease
  - Some cases of odontogenic sinusitis
- Dental treatment with antibiotics
  - Odontogenic sinusitis

Pneumonia

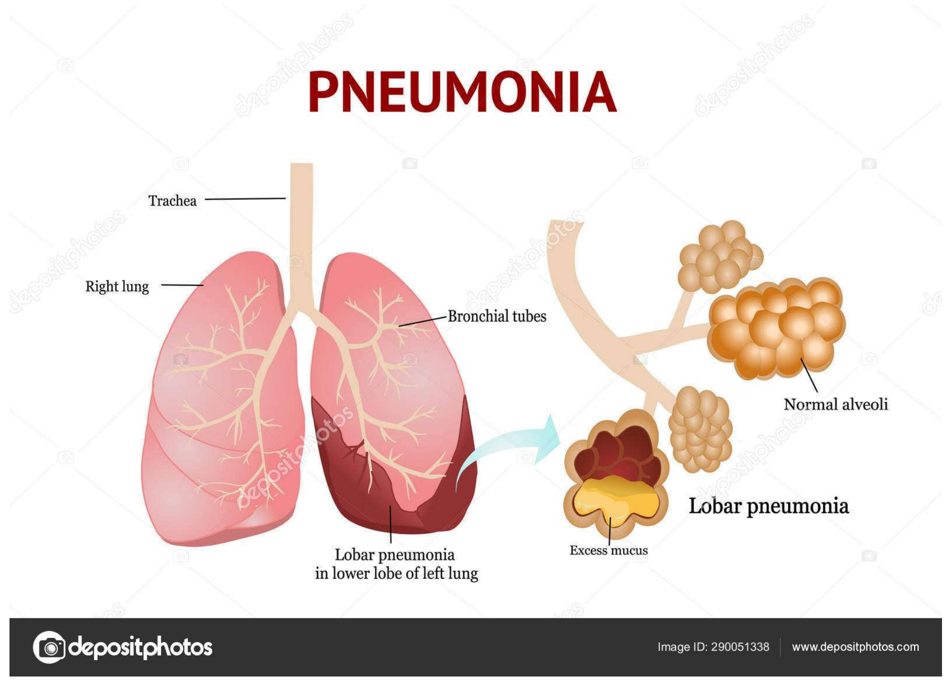
Type of Infection	Microbial Organism
Bacterial	*Streptococcus Pneumoniae, Hemophilus Influenzae, Staphylococcus Aureus* *Mycoplasma Pneumoniae, Chlamydophila Pneumoniae, Legionella Pneumoniae* *Mycobacteria Tuberculosis*
Viral	*Influenza, SARS-CoV-2*
Fungal	*Coccidioidomycosis, Histoplasmosis, Blastomycosis, Cryptococcus, Pneumocystis Jiroveci (seen in immunocompromised patients- HIV)*

Location in Lungs
Bronchopneumonia
Atypical/Interstitial Pneumonia
Lobar Pneumonia

How Disease is Acquired
Community Acquired Pneumonia
Nosocomial Pneumonia Ventilator
Aspiration Pneumonia

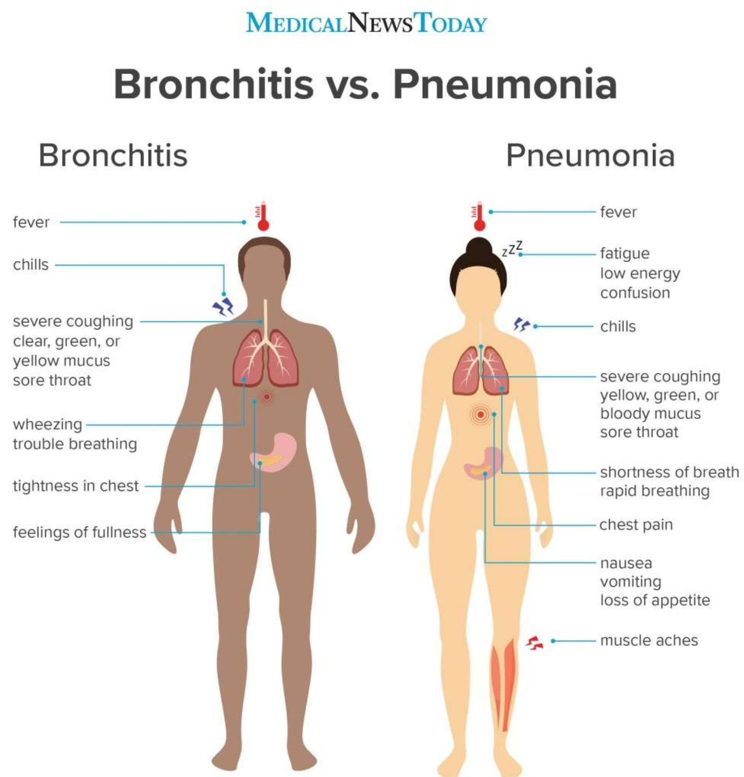
Lobar Pneumonia

1. **Congestion**
   - Fluid accumulation associated with inflammation due to microbial invasion of the alveoli.
   - Lung becomes heavy and boggy
2. **Red hepatization**
   - Red blood cells, neutrophils, and fibrin fill the alveolar sacs
   - Lung tissue becomes more solid
3. **Grey hepatization**
   - Red blood cells disintegrate, but the lung tissue remains firm
4. **Resolution**
   - Exudate is digested and cough out of the lung (productive cough)



Symptoms of Pneumonia

- Productive Cough
- Sharp chest pain
- Dyspnea
- Tachypnea
- Chills
- Fever
- Muscle pain



Prevention of Aspiration Pneumonia in Susceptible Patients

- Standardized clinical assessment of oral condition
- Use of a soft-bristle brush
- Use of mouth swabs when and where brushing is contraindicated
- Do not use tap water for oral hygiene
- Subglottic suctioning in mechanical ventilators
- Clean storage of all oral hygiene tools

Tuberculosis (TB)

- Infectious disease caused by *Mycobacterium tuberculosis*
- The most common site of infection is the lung
- Latent TB Infection (LTBI)
- Active TB Infection
- TB infection of other organs

Diagnosis

- Mantoux purified protein derivative (PPD)
- Chest X-ray
- Sputum culture
- QuantiFERON Gold blood test

TB Screening in the Dental Office

- Dentists and dental staff can be exposed to TB during patient treatment.
- Office protocol for encounters with active patients.
- Healthcare providers should receive regular TB screenings.

Dental management of patients with history of tuberculosis (TB)

Patients with active TB disease
- Palliative care with medications if hospital setting is not available.
- Any urgent care involving aerosols must be done in an isolation setting with negative pressure ventilation and appropriate personal respiratory protection, often in a hospital setting

Patients with symptoms suggestive of TB disease
- Refer immediately to physician for evaluation
- If coughing, give patient a surgical mask and place in an isolated area until transportation can be arranged

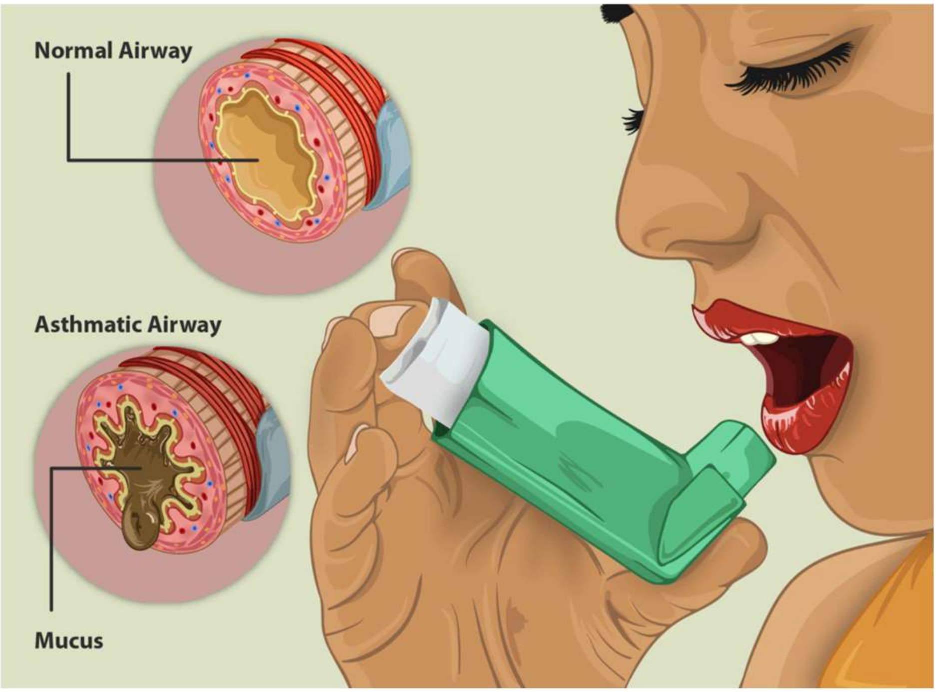
Patient with past history of TB disease
- Careful medical history and review of systems (ROS)
- Establish that patient has been adequately treated and has had negative sputum cultures and chest X-ray demonstrating patient is noninfectious
- Consult with physician if follow-up evaluation is questionable or symptoms of active disease present
- Provide routine dental care if there has been appropriate medical follow-up and no signs of clinically active disease

Patient with history of LTBI
- Medical history and ROS
- Verify medical evaluation to rule out active disease
- Determine that patient received prophylactic therapy with isoniazid for at least 6 months
- Treat as routine patient

The ADA Practical Guide to Patients with Medical Conditions

Asthma

- Inflammatory disorder of the upper airway
  - Hyperresponsive reaction to extrinsic and intrinsic stimuli
- Clinical presentation
  - Tightness of the chest, wheezing, and breathlessness
  - Mild to Moderate disease
  - Status asthmaticus



www.leogenic.com/asthma-causes-symptoms-types-diagnosis-treatment-prevention/

Asthma

- Dental Considerations
  - What triggers the patient's asthma?
  - When did the last asthma attack occur?
  - How often do asthma attacks occur?
  - Hospitalizations associated with asthma
  - Medications for asthma management
  - Have the patient bring their rescue inhaler to every appointment

Asthma

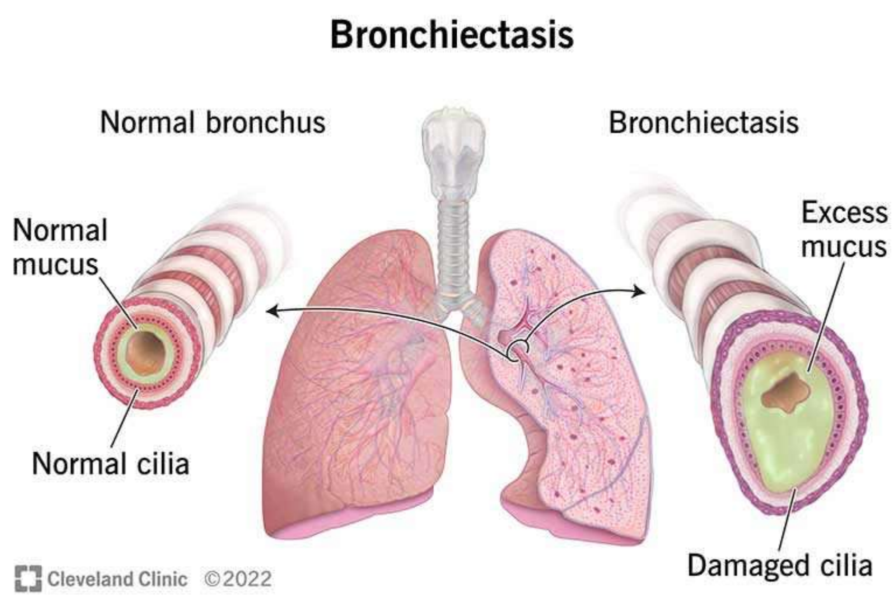
- Dental Emergency
  - Cessation of all treatment
  - Remove all dental instruments from the oral cavity
  - Raise the dental chair
  - Utilize the rescue inhaler and administer oxygen
  - Cutaneous injections of epinephrine
  - Contract emergency services

Asthma

- Treatment
  - Non-pharmacological
    - Lifestyle modifications
  - Pharmacological
    - Be mindful of drug interactions
    - Xerostomia
      - Caries
      - Fungal infections
  - Increased oral hygiene efforts

Bronchiectasis

- Disorder presenting with damage to the bronchi
  - Dilation and distortion of the bronchi
  - Excessive mucus
  - Loss of cilia and epithelial cells
  - Scarring of the bronchial tissue
  - Bacterial lung infections
- Clinical presentation
  - Productive cough
  - Hemoptysis
  - Clubbing of fingers

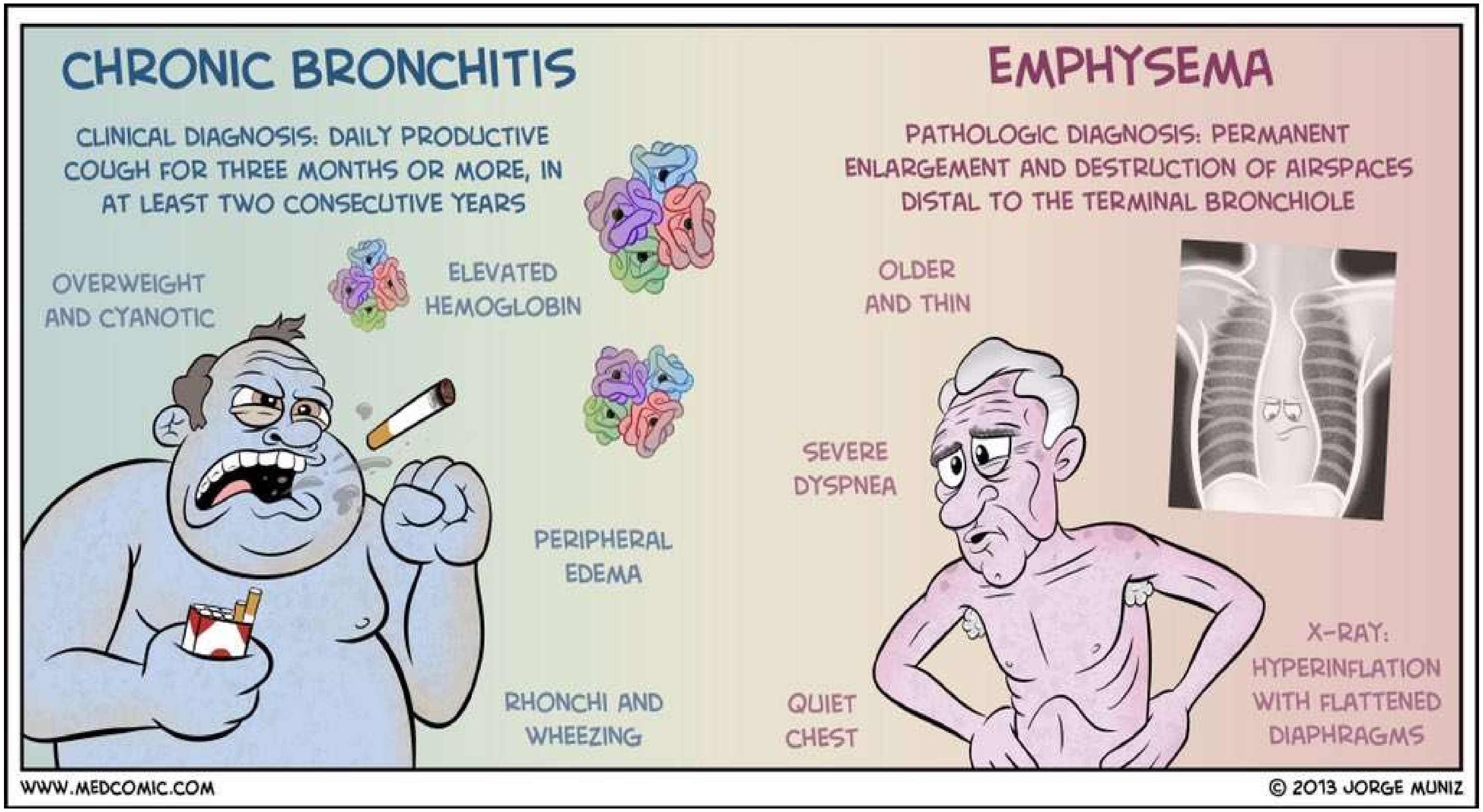


Acute Bronchitis

- Inflammation of the large airways (trachea & bronchi)
- Short duration (up to 3 weeks)
- Viral
  - Productive or non-productive cough, chest discomfort, and dyspnea
- Bacterial
  - Productive cough (purulent)
  - Chest pain
- Chest imaging utilized to distinguish between bronchitis and pneumonia
  - Bronchitis effects airways while pneumonia occurs in the lung parenchyma

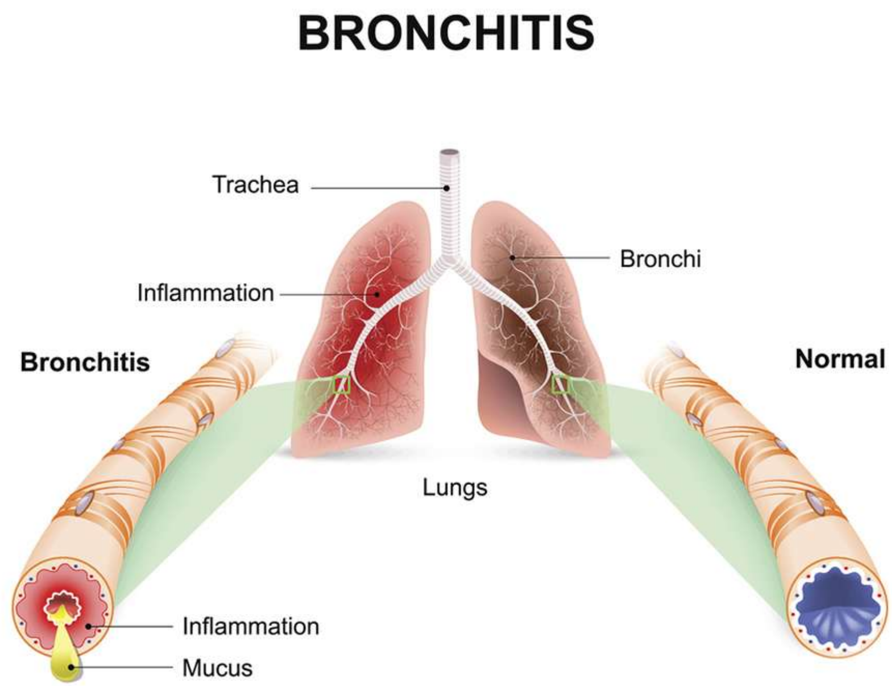
Chronic Obstructive Pulmonary Disease (COPD)

1. Bronchitis
2. Emphysema



Chronic Bronchitis

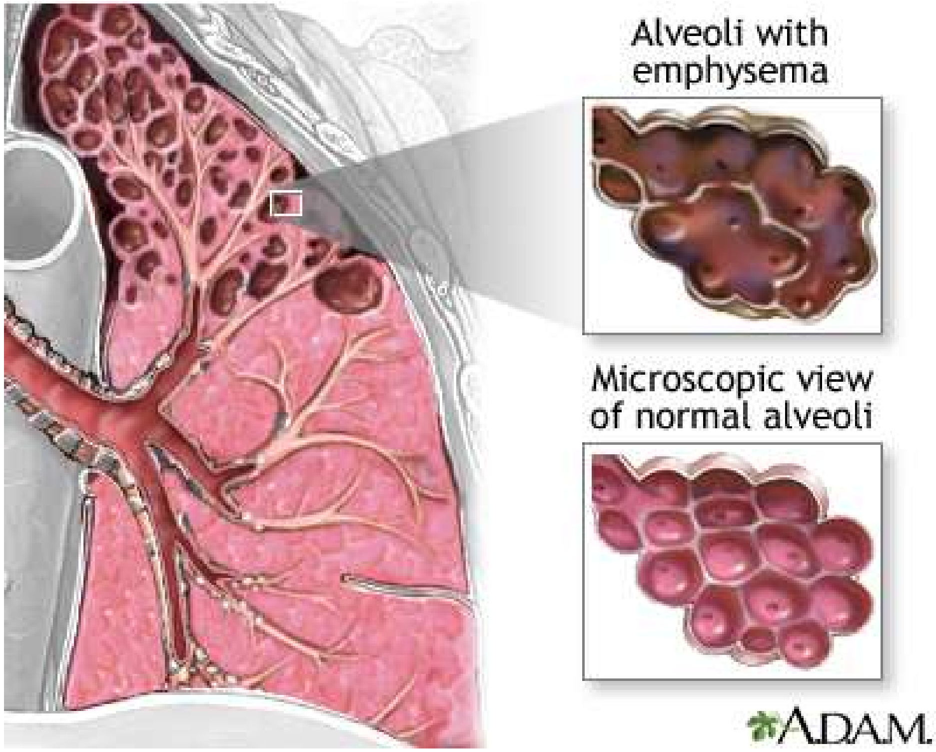
- Airway obstruction due to chronic inflammation of the bronchioles
- Clinical Presentation
  - Productive cough
  - Progressive dyspnea
  - Blue discoloration of the skin
  - Fatigue
  - Wheezing
  - Weight gain



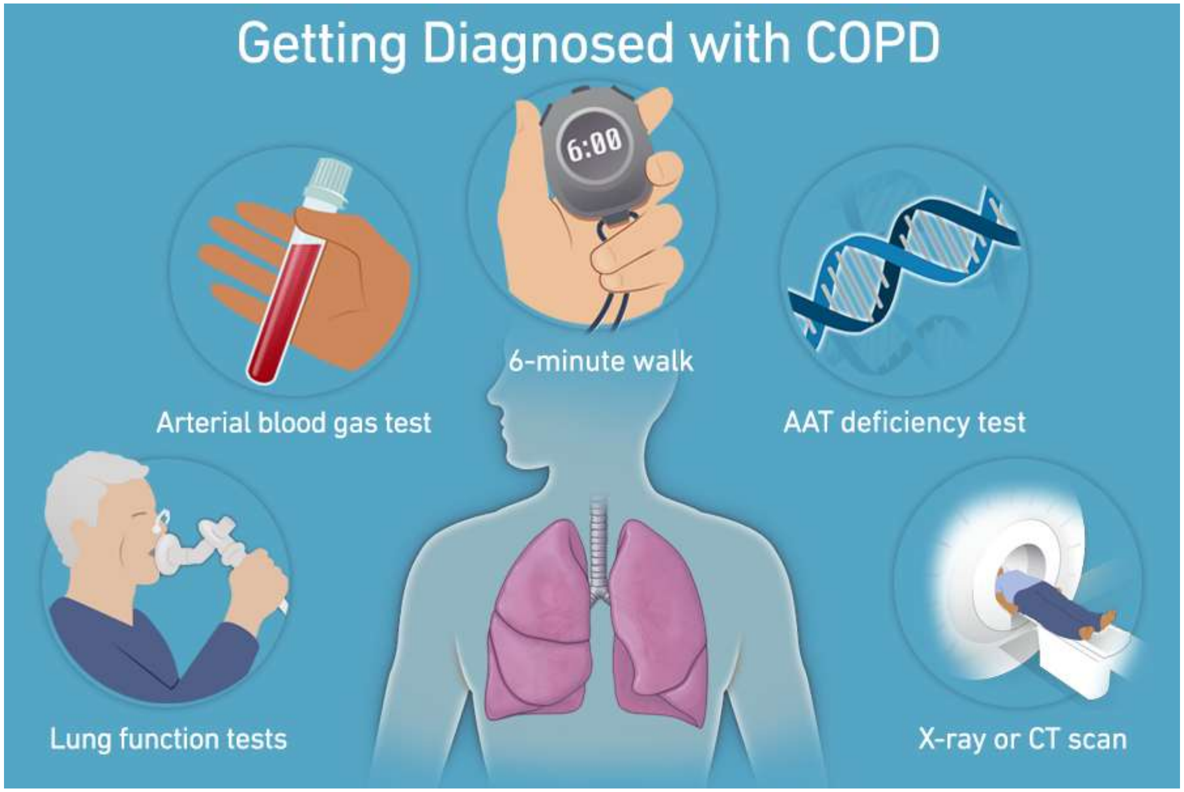
www.cdc.gov/antibiotic-use/bronchitis.html

Emphysema

- Irreversible enlargement of bronchioles and alveoli
  - **Centriancinar (Centrilobar) emphysema**
  - **Panacinar (Panlobar) emphysema**
- Clinical presentation
  - Progressive dyspnea
  - Enlargement of the chest wall
  - Weight loss
  - Pink discoloration of the skin



Diagnosis of COPD



Chronic Obstructive Pulmonary Disease

- Dental considerations
  - Disease severity
    - Pulse oximetry < 91%- DON'T TREAT
  - Susceptibility to infection
  - Risk associated with nitrous oxide administration
  - Drug interaction and adverse reactions
- Treatment
  - Prevention
  - Drug therapy
  - Lung transplant

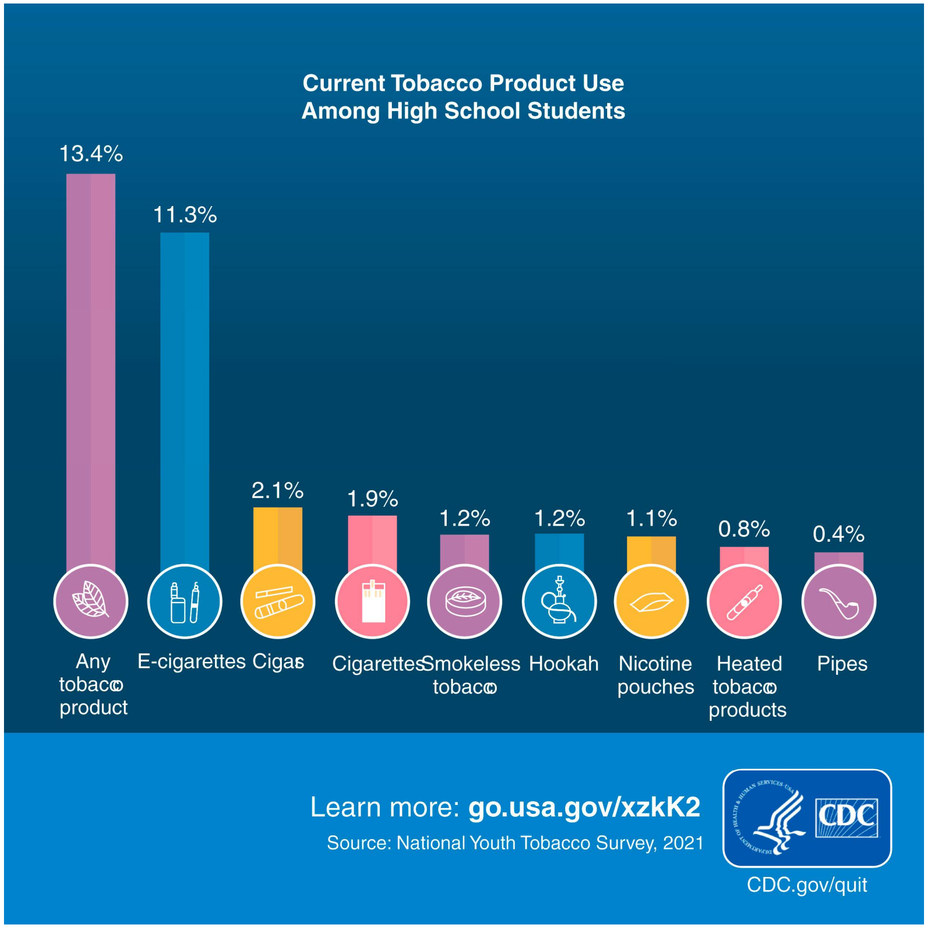
Lung Cancer

- Second most common cause of cancer related deaths in the United States
- Risk factors: smoking and exposure to man-made mineral fibers (silica and asbestos)
- Symptoms
  - Coughing, hemoptysis, wheezing, chest pain, anorexia, and fatigue
- Dental care prior to, during, and after cancer therapy

Smoking and Tobacco Cessation

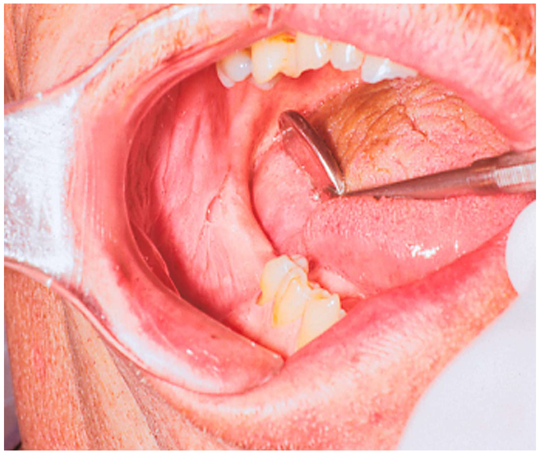
- Physiologic effects of nicotine
  - **Affects dopaminergic pathways**
  - Increase in cardiac output, heart rate, and blood pressure
  - Decrease in appetite

- Sign of withdrawal
  - Cravings
  - Agitation
  - Restlessness
  - Anxiety
  - Hunger
  - Insomnia
  - Lack of focus



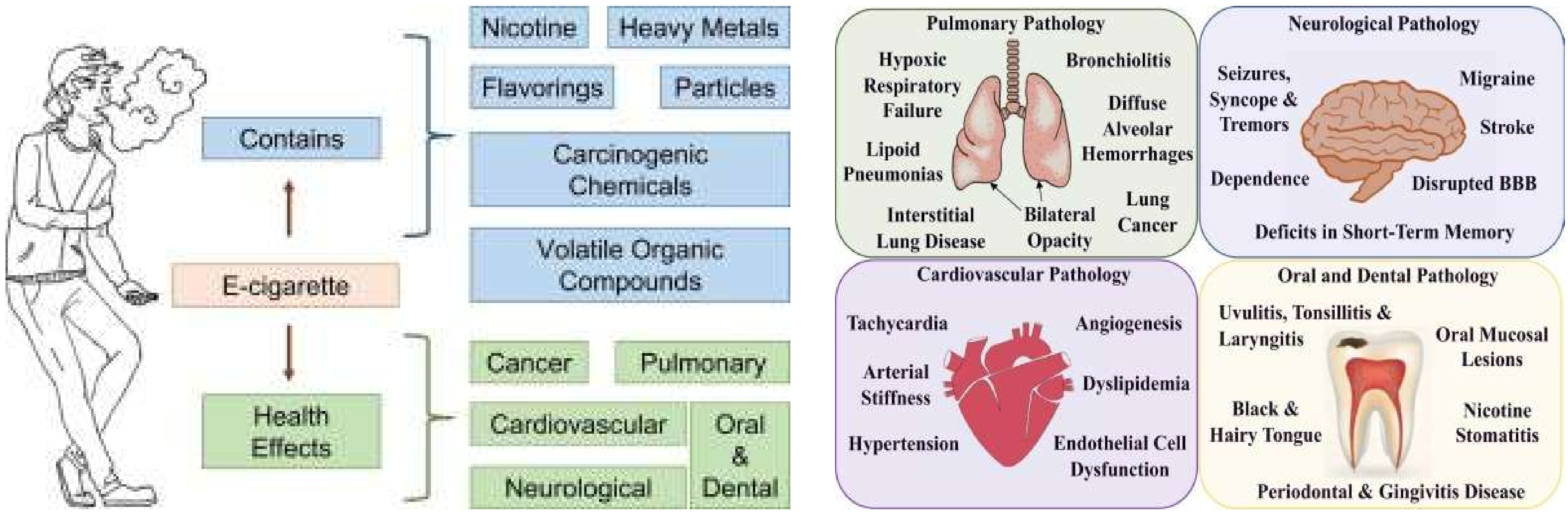
Smoking and Tobacco Cessation

- Oral manifestations
  - **Xerostomia**
  - **Halitosis**
  - **Staining of the teeth**
  - Hairy tongue
  - Increased pigmentation “Smokers Melanosis”
  - **Increased risk of gingivitis and periodontal disease**
  - Nicotinic stomatitis
  - Smokeless tobacco keratosis (smokeless tobacco)
  - **Premalignant lesions/oral cancer**



Figures 8.5 & 8.9 Little and Falace’s Management of the Medically Compromised Patient 10th Ed.

Electronic Cigarettes



Toxicology Reports Volume 9, 2022, Pages 1357-1368

Smoking Cessation

- Smoking cessation discussed by the dentists
  - Take an assessment to determine the nicotine dependency of the patient.
  - Discuss the oral side effects associated with tobacco
  - Present different smoking cessation options
    - Interprofessional programs
    - Prescription and suggestion of OTC medications for cessation
    - Refer patient for counseling

Telephone Help and Quit Lines

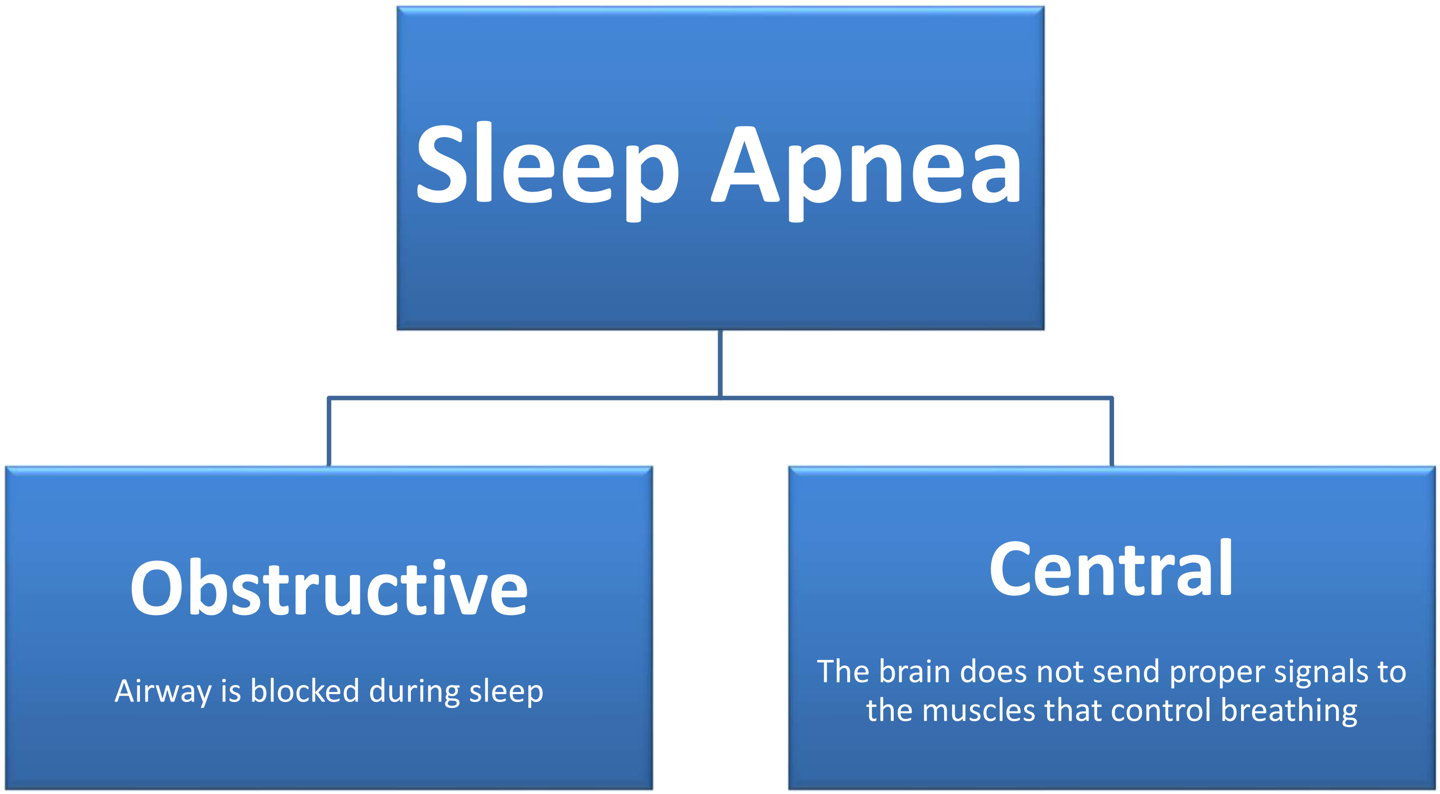
- 800-QUITNOW (US Department of Health and Human Services national quit line)
- 877-44-U-QUIT (National Cancer Institute dedicated quit smoking line)
- 877-YES-QUIT (American Cancer Society quit line)
- 800-4-CANCER (Cancer Information Service of the National Cancer Institute)

Helpful Websites

- www.surgeongeneral.gov/tobacco
- www.smokefree.gov
- www.nlm.nih.gov/medlineplus/smokingcessation.html
- www.cancer.gov/cancertopics/pdq/prevention/control-of-tobacco-use/HealthProfessional
- www.cdc.gov/tobacco
- www.cancer.org/docroot/PED/content/PED_10_13X_Guide_for_Quitting_Smoking.asp

Box 8.2-Resources for Support Material
Little and Falace's Management of the Medically Compromised Patient 10th Ed.

Sleep Apnea


Sleep Apnea
Obstructive
Airway is blocked during sleep
Central
The brain does not send proper signals to the muscles that control breathing

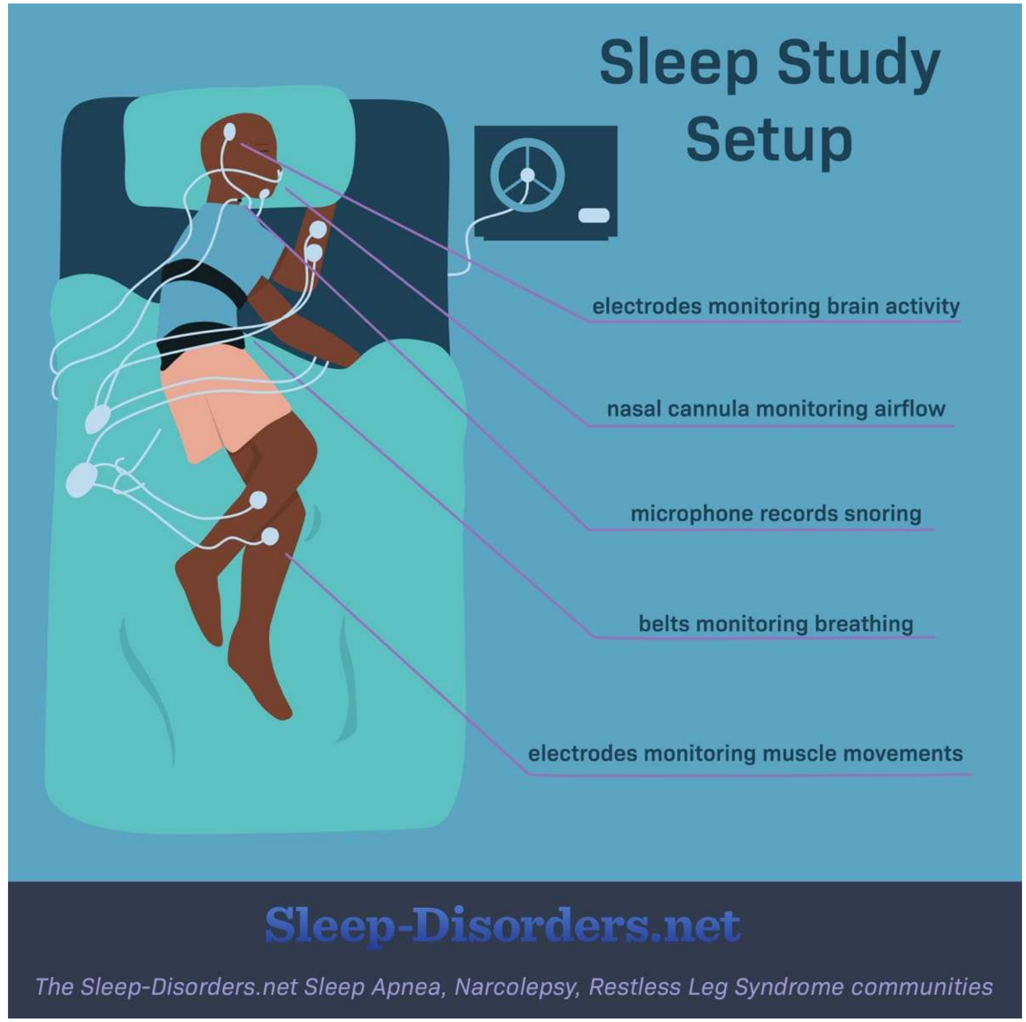
Obstructive Sleep Apnea Syndrome

- Causes- narrowing of the upper airway or positioning of the tongue, soft, and muscles adjacent to the airway.
- Classifications
  - Mild, moderate, and severe
- Risk factors
  - Obesity, smoking, alcoholism, hypertension, anatomical deviations of the nasal cavity, snoring, enlargement of the muscles of the soft palate, retrognathic mandible

Obstructive Sleep Apnea Syndrome

Polysomnography

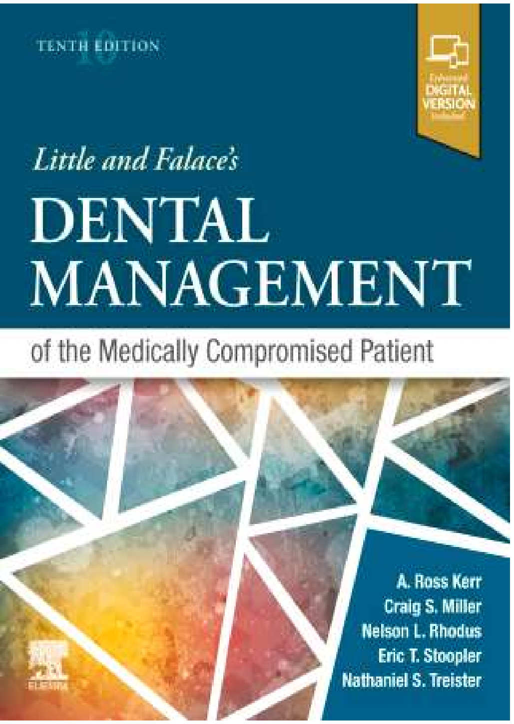
- A test that measures and records brain activity, breathing, and movement during sleep.
- Sleeps studies are conducted in a laboratory setting.
- The sleep study must be requested by the patient’s physician.
- Treatment options are based on the diagnosis obtained from the sleep study.

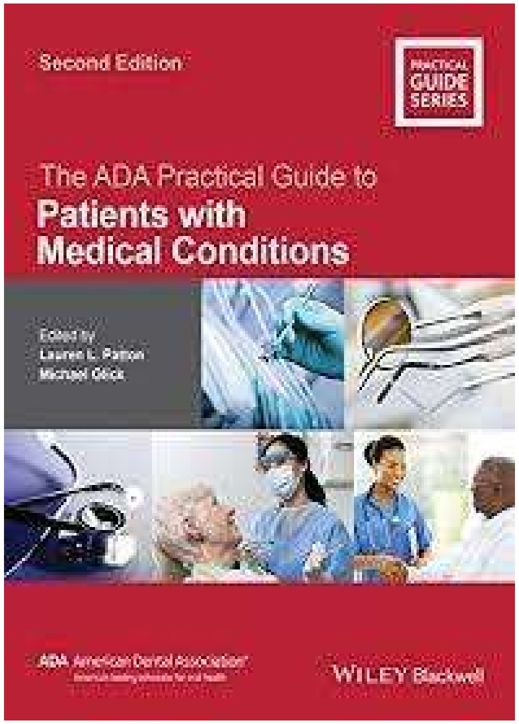


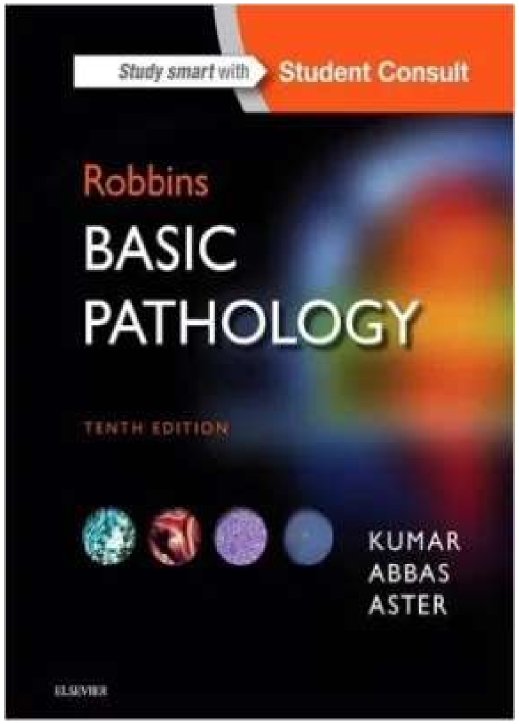
Treatment of Sleep Apnea

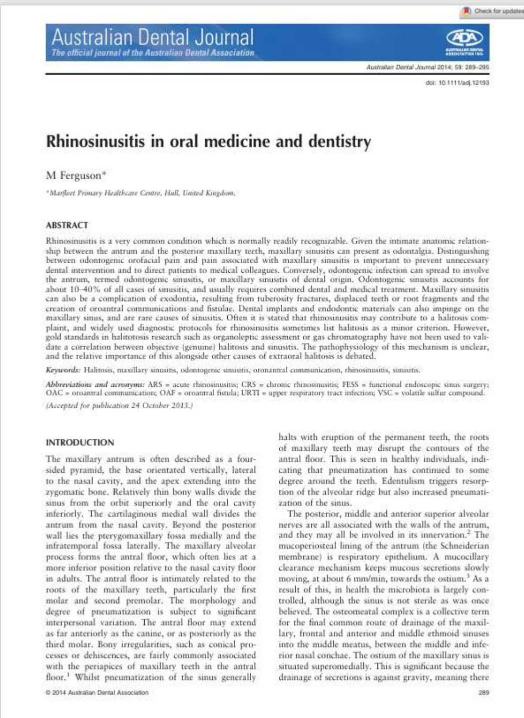
- Positive Airway Pressure
  - Continuous Positive Airway Pressure (CPAP) machine
- Oral Appliances
  - **Mandibular Advancement Devices**
  - **Tongue Retaining Devices**
- Surgery
  - Uvulopalatopharyngoplasty (UPPP)
  - Laser-assisted uvulopalatoplasty (LAUP)
  - Radiofrequency volumetric tissue reduction (RVTR)
  - Genioglossus advancement-hyoid myotomy and suspension (GAHMS)
  - Maxillary and mandibular advancement osteotomy (MMO)

Resources









Check for updates

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Rhinosinusitis in oral medicine and dentistry

M Ferguson*
*Marfleet Primary Healthcare Centre, Hull, United Kingdom.

ABSTRACT

Rhinosinusitis is a very common condition which is normally readily recognizable. Given the intimate anatomic relationship between the antrum and the posterior maxillary teeth, maxillary sinusitis can present as odontalgia. Distinguishing between odontogenic orofacial pain and pain associated with maxillary sinusitis is important to prevent unnecessary dental intervention and to direct patients to medical colleagues. Conversely, odontogenic infection can spread to involve the antrum, termed odontogenic sinusitis, or maxillary sinusitis of dental origin. Odontogenic sinusitis accounts for about 10–40% of all cases of sinusitis, and usually requires combined dental and medical treatment. Maxillary sinusitis can also be a complication of exodontia, resulting from tuberosity fractures, displaced teeth or root fragments and the creation of oroantral communications and fistulae. Dental implants and endodontic materials can also impinge on the maxillary sinus, and are rare causes of sinusitis. Often it is stated that rhinosinusitis may contribute to a halitosis complaint, and widely used diagnostic protocols for rhinosinusitis sometimes list halitosis as a minor criterion. However, gold standards in halitotosis research such as organoleptic assessment or gas chromatography have not been used to validate a correlation between objective (genuine) halitosis and sinusitis. The pathophysiology of this mechanism is unclear, and the relative importance of this alongside other causes of extraoral halitosis is debated.

Keywords: Halitosis, maxillary sinusitis, odontogenic sinusitis, oronantral communication, rhinosinusitis, sinusitis.

Abbreviations and acronyms: ARS = acute rhinosinusitis; CRS = chronic rhinosinusitis; FESS = functional endoscopic sinus surgery; OAC = oroantral communication; OAF = oroantral fistula; URTI = upper respiratory tract infection; VSC = volatile sulfur compound.

(Accepted for publication 24 October 2013.)

INTRODUCTION

The maxillary antrum is often described as a four-sided pyramid, the base orientated vertically, lateral to the nasal cavity, and the apex extending into the zygomatic bone. Relatively thin bony walls divide the sinus from the orbit superiorly and the oral cavity inferiorly. The cartilaginous medial wall divides the antrum from the nasal cavity. Beyond the posterior wall lies the pterygomaxillary fossa medially and the infratemporal fossa laterally. The maxillary alveolar process forms the antral floor, which often lies at a more inferior position relative to the nasal cavity floor in adults. The antral floor is intimately related to the roots of the maxillary teeth, particularly the first molar and second premolar. The morphology and degree of pneumatization is subject to significant interpersonal variation. The antral floor may extend as far anteriorly as the canine, or as posteriorly as the third molar. Bony irregularities, such as conical processes or dehiscences, are fairly commonly associated with the periapices of maxillary teeth in the antral floor.[1] Whilst pneumatization of the sinus generally halts with eruption of the permanent teeth, the roots of maxillary teeth may disrupt the contours of the antral floor. This is seen in healthy individuals, indicating that pneumatization has continued to some degree around the teeth. Edentulism triggers resorption of the alveolar ridge but also increased pneumatization of the sinus.

The posterior, middle and anterior superior alveolar nerves are all associated with the walls of the antrum, and they may all be involved in its innervation.[2] The mucoperiosteal lining of the antrum (the Schneiderian membrane) is respiratory epithelium. A mucocillary clearance mechanism keeps mucous secretions slowly moving, at about 6 mm/min, towards the ostium.[3] As a result of this, in health the microbiota is largely controlled, although the sinus is not sterile as was once believed. The osteomeatal complex is a collective term for the final common route of drainage of the maxillary, frontal and anterior and middle ethmoid sinuses into the middle meatus, between the middle and inferior nasal conchae. The ostium of the maxillary sinus is situated superomedially. This is significant because the drainage of secretions is against gravity, meaning there